

①
★ Squeeze theorem: [Thm: 1.6.4, H.A, page-123]

let $f(x)$, $g(x)$, $h(x)$ be functions satisfying

$$g(x) \leq f(x) \leq h(x).$$

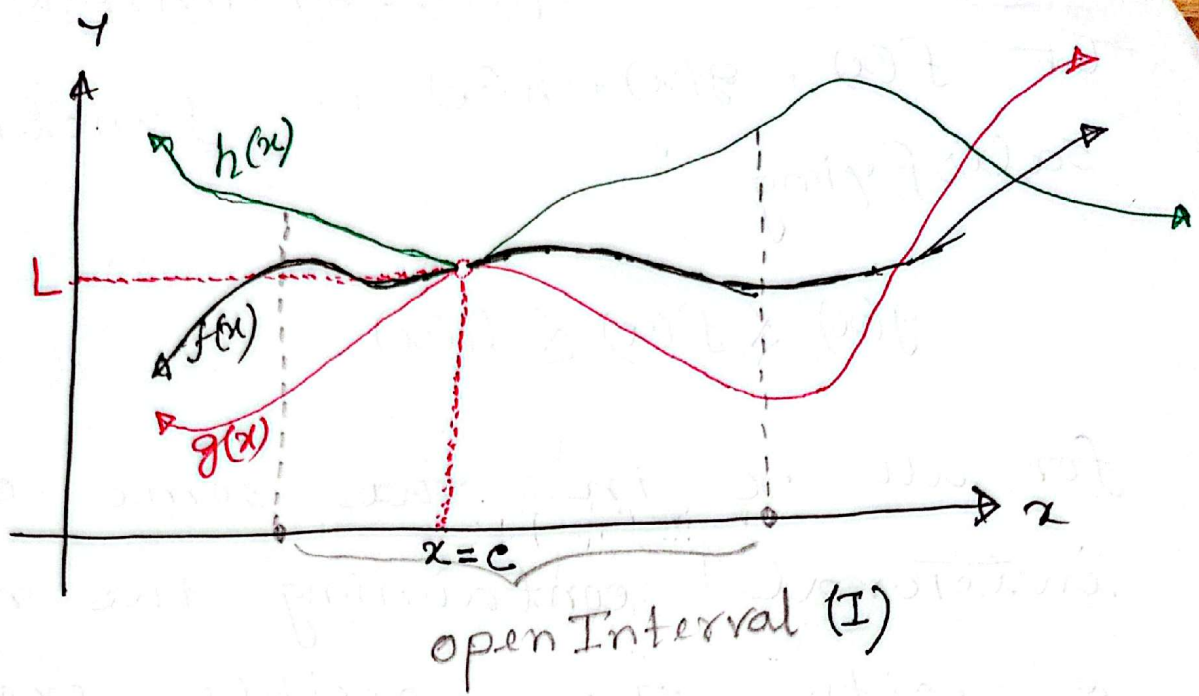
for all x in ~~the~~ some open interval containing the number c , with the possible exception that the inequalities need not hold at c .

If $g(x)$ and $h(x)$ have the same limit as x approaches c ,

$$\text{say } \lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$$

then $f(x)$ also has this limit as x approaches to c , that is

$$\lim_{x \rightarrow c} f(x) = L.$$



$$g(x) \leq f(x) \leq h(x), \forall x \in I.$$

$$\Rightarrow \lim_{x \rightarrow c} g(x) \leq \lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} h(x)$$

$$\Rightarrow L \leq \lim_{x \rightarrow c} f(x) \leq L$$

$$\Rightarrow \lim_{x \rightarrow c} f(x) = L.$$

* Find $\lim_{x \rightarrow 0} x \cdot \sin\left(\frac{1}{x}\right)$ using squeezing theorem.

Ans

we know,

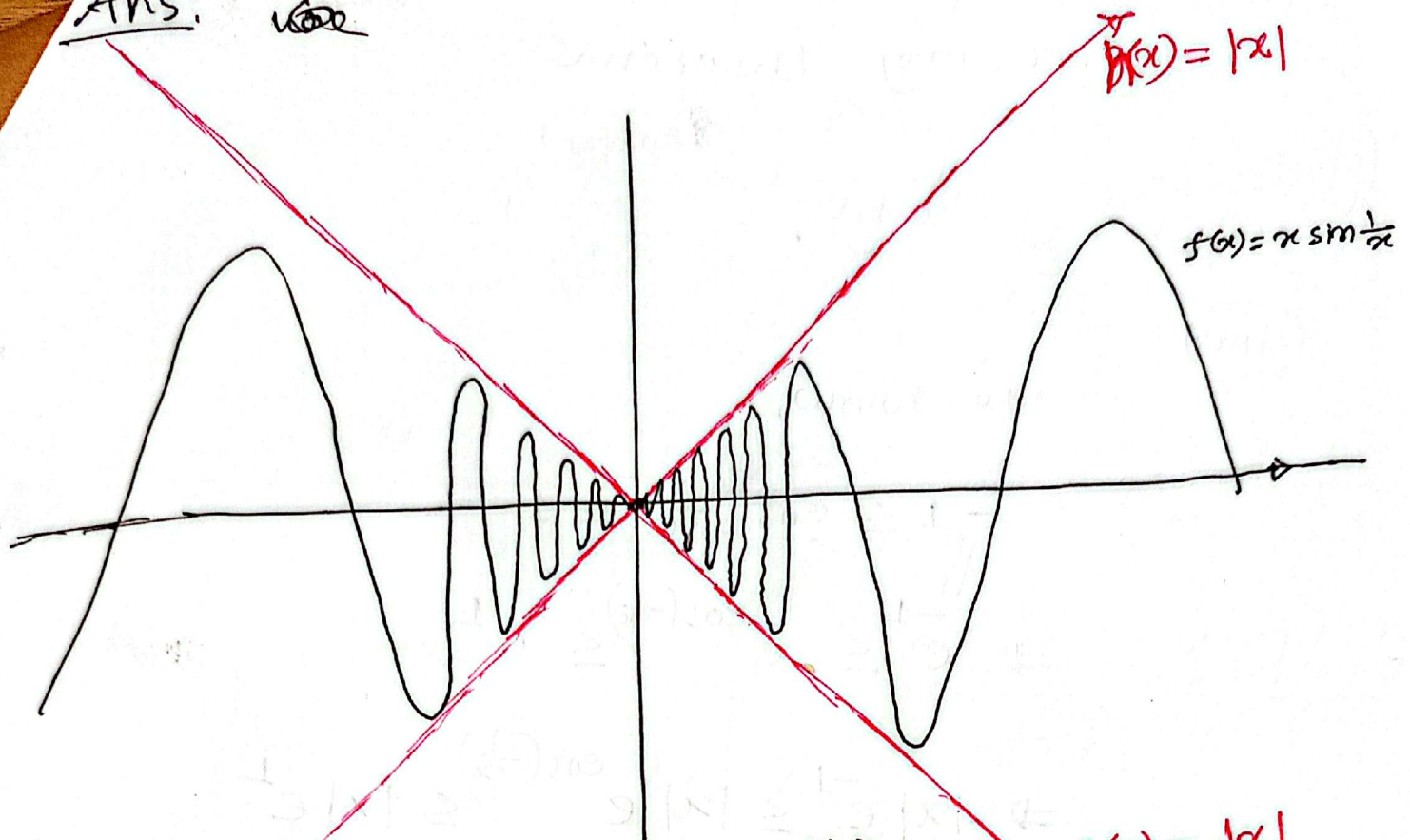
$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

$$\Rightarrow -1 \times |x| \leq |x| \sin\left(\frac{1}{x}\right) \leq 1 \times |x|$$

$$\Rightarrow -|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$$

* Find $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$ using squeeze theorem. (3)

Ans: ~~use~~



we know, $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$
 $\Rightarrow -|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$

$$\Rightarrow \lim_{x \rightarrow 0} -|x| \leq \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) \leq \lim_{x \rightarrow 0} |x|$$

$$\Rightarrow 0 \leq \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) \leq 0$$

$$\therefore \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$$

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★ Evaluate the limit using squeezing theorem

$$\lim_{x \rightarrow 0} x e^{\cos(\frac{1}{x})}$$

Ans:

we know,

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$$

$$\Rightarrow e^{-1} \leq e^{\cos(\frac{1}{x})} \leq e^1$$

$$\Rightarrow -|x| e^{-1} \leq |x| e^{\cos(\frac{1}{x})} \leq |x| e^1$$

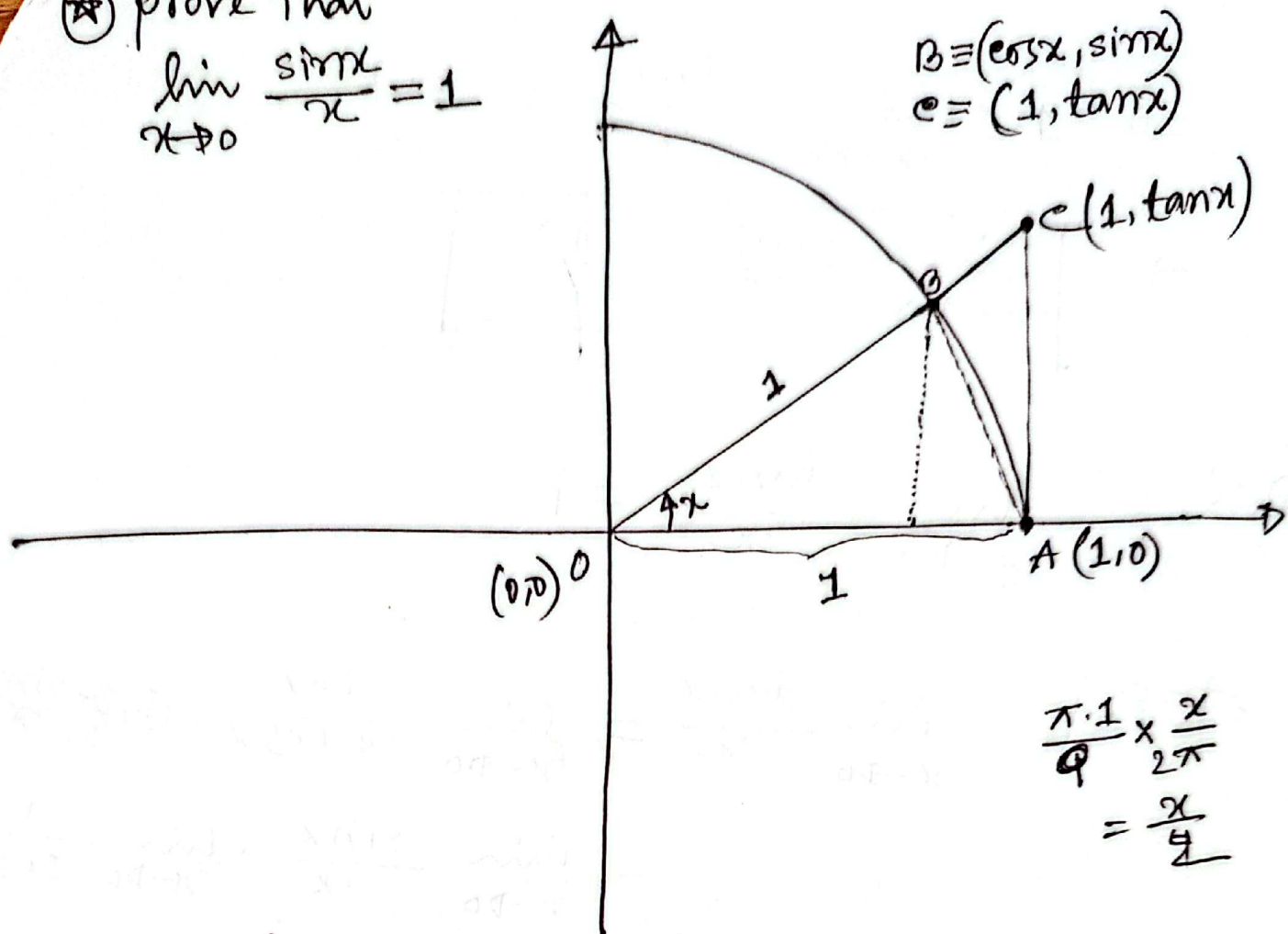
$$\Rightarrow -|x| e^{-1} \leq x e^{\cos(\frac{1}{x})} \leq |x| e^1$$

$$\Rightarrow \lim_{x \rightarrow 0} -|x| e^{-1} \leq \lim_{x \rightarrow 0} x e^{\cos(\frac{1}{x})} \leq \lim_{x \rightarrow 0} |x| e^1$$

$$\Rightarrow 0 \leq \lim_{x \rightarrow 0} x e^{\cos(\frac{1}{x})} \leq 0$$

$$\therefore \lim_{x \rightarrow 0} x e^{\cos(\frac{1}{x})} = 0$$

⑤ prove that
 $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$



Ans:

Area of triangle OAB $\leq$ Area of sector OAB $\leq$ Area of triangle OAC

$$\Rightarrow \frac{1}{2} \times 1 \times \sin x \leq \frac{x}{2} \leq \frac{1}{2} \times 1 \times \tan x$$

$$\Rightarrow \sin x \leq x \leq \tan x$$

$$\Rightarrow \cancel{\cos x} \quad 1 \leq \frac{x}{\sin x} \leq \frac{1}{\cos x}$$

$$\Rightarrow \cos x \leq \frac{\sin x}{x} \leq 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \cos x \leq \lim_{x \rightarrow 0} \frac{\sin x}{x} \leq \lim_{x \rightarrow 0} 1$$

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$$\Rightarrow 1 \leq \lim_{x \rightarrow 0} \frac{\sin x}{x} \leq 1$$

$$\Rightarrow \boxed{\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1}$$

Ex. $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$

Ans. $\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x \cos x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x}$

$$= 1 \cdot \frac{1}{\cos 0} = 1$$

$$\therefore \boxed{\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1}$$

Ex. $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x} = \frac{3}{5}$

Ans. $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x} = \lim_{x \rightarrow 0} \frac{\frac{\sin 3x}{3x}}{\frac{\sin 5x}{5x}} \times \frac{3}{5}$

$$= \frac{\lim_{x \rightarrow 0} \frac{\sin 3x}{3x}}{\lim_{x \rightarrow 0} \frac{\sin 5x}{5x}} \times \frac{3}{5} = \frac{3}{5} \quad \text{Ans.}$$

⊛ Ex.

show that

(7)

(a) $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

but

(b) $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$ does exist.

Ans.

(a)

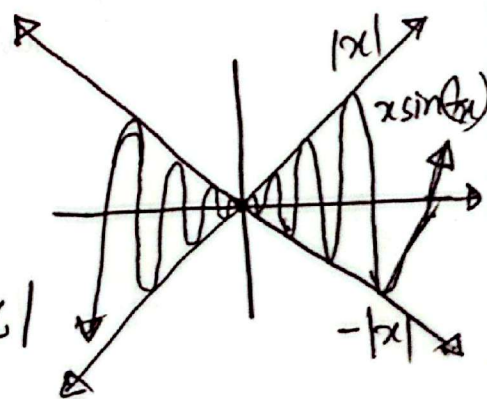
see from H.A Book page: 125
 $\lim_{x \rightarrow 0^+} \sin\left(\frac{1}{x}\right)$, ~~where~~ oscillates between
-1 to 1 and similarly

$\lim_{x \rightarrow 0^-} \sin\left(\frac{1}{x}\right)$ also oscillates
between -1 to 1. so

$\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist.

(b) $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$

$\Rightarrow -|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$



$\Rightarrow \lim_{x \rightarrow 0} -|x| \leq \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) \leq \lim_{x \rightarrow 0} |x|$

$\Rightarrow 0 \leq \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) \leq 0$

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$$\therefore \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0.$$

Ex. Test the continuity of the function

$$f(x) = \begin{cases} x \cos\left(\frac{1}{x}\right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

at $x = 0$.

Ans: for continuity at $x=0$ if $f(x)$.

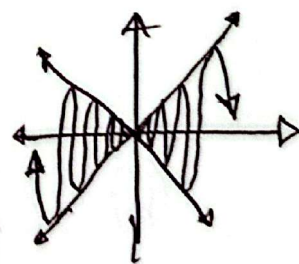
$$(1) f(0) = 0$$

$$(2) \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$$

way 1 to calculate this limit
we know,

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$$

$$\Rightarrow -|x| \leq x \cos\left(\frac{1}{x}\right) \leq |x|$$



$$\Rightarrow \lim_{x \rightarrow 0} (-|x|) \leq \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) \leq \lim_{x \rightarrow 0} |x|$$

$$\Rightarrow 0 \leq \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) \leq 0$$

$$\therefore \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$$

$$(3) \quad f(0) = \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$$

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So, $f(x)$ is continuous at $x=0$.

Way two to calculate $\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$

we know,

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$$

if $x > 0$, then

$$-x \leq x \cos\left(\frac{1}{x}\right) \leq x$$

$$\Rightarrow \lim_{x \rightarrow 0^+} (-x) \leq \lim_{x \rightarrow 0^+} x \cos\left(\frac{1}{x}\right) \leq \lim_{x \rightarrow 0^+} x$$

$$\Rightarrow 0 \leq \lim_{x \rightarrow 0^+} x \cos\left(\frac{1}{x}\right) \leq 0$$

$$\Rightarrow \lim_{x \rightarrow 0^+} x \cos\left(\frac{1}{x}\right) = 0 \quad [\text{R.H.L}]$$

If $x < 0$, then

$$-x \geq x \cos\left(\frac{1}{x}\right) \geq x$$

$$\Rightarrow \lim_{x \rightarrow 0^-} (-x) \geq \lim_{x \rightarrow 0^-} x \cos\left(\frac{1}{x}\right) \geq \lim_{x \rightarrow 0^-} x$$

$$\Rightarrow \textcircled{0} \geq \lim_{x \rightarrow 0^-} x \cos\left(\frac{1}{x}\right) \geq 0$$

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$$\therefore \lim_{x \rightarrow 0^-} x \cos\left(\frac{1}{x}\right) = 0 \quad [L.H.L.]$$

$$\text{Here } \lim_{x \rightarrow 0^-} x \cos\left(\frac{1}{x}\right) = 0 = \lim_{x \rightarrow 0^+} x \cos\left(\frac{1}{x}\right)$$

$$\Rightarrow \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$$